EXERCISE 4 25/10/2024

Question 1

SQL QUERY

```
SELECT account_id,
       account_holder,
       City
FROM branch_station_accounts
UNION
SELECT account_id,
       account_holder,
       City
FROM branch_rosebank_accounts;
```

Results Table

accounts_id	account_holder	City
A001	Nomvula Dlamini	NULL
A002	David Maseng	NULL
A003	Lerato Sithole	NULL
A004	Peter Nkosi	NULL
A005	Zanele Khumalo	NULL

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Question 3

SQL Query

```
SELECT product_code,
       product_name,
       'Savings' AS product_type
FROM savings_products
UNION
SELECT product_code,
       product_name,
       'Current' AS product_type
FROM current_products;
```

Results table

product_code	product_name	product_type
SV01	Basic Savings	Savings
SV02	Premium Savings	Savings
SV03	Youth Savings	Savings
SV04	Business Savings	Savings
CR01	Standard Current	Current
CR02	Gold Current	Current
C03	Business Current	Current

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Question 3

SQL Query

```
SELECT  staff_id,
        staff_name,
        email
FROM    retail_banking_staff
UNION
SELECT  staff_id,
        staff_name
        email
FROM    corporate_banking_staff;
```

Results Table

staff_id	staff_name	email
S001	Mpho Radebe	Null
S002	Brian Tshabalala	Null
S003	Aisha Patel	Null
S004	Kabelo Mosiolo	Null
S005	Nandi Dube	Null
S006	Sipho Khumalo	Null

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Question 4

SQL Query

```
SELECT city_code,
       city_name,
       region
FROM mobile_branch_cities
Union
SELECT city_code,
       city_name,
       region
FROM digital_branch_cities;
```

Results Table

City_code	City_name	region
C01	Johannesburg	Null
C02	Pretoria	Null
C03	Cape Town	Null
C04	Durban	Null
C05	Polokwane	Null
C06	Port Elizabeth	Null

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Question 5

SQL Query

```
SELECT customer_id,
       Customer_name,
       Segment
FROM push_notification_targets
UNION
SELECT Customer_id,
       Customer_name,
       Segment
FROM inapp_banner_targets;
```

Results Table

Customer_id	Customer_name	Segment
C1001	Nomsa Zuma	Null
C1002	Andile Buthelezi	Null
C1003	Fatima Mohamed	Null
C1004	Ryno van Zyl	Null
C1005	Thandeka Cele	Null
C1006	Samuel Nkosi	Null

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Question 6

SQL Query

```
SELECT txn_id AS transaction_id,
       account_id,
       amount
FROM atm01_transactions
UNION ALL
SELECT txn_id AS transaction_id,
       account_id,
       amount
FROM atm02_transactions;
```

Result Table

transaction_id	account_id	amount	transaction_date
T1001	A001	500.00	Null
T1002	A002	1209.00	Null
T1003	A001	300.00	Null
T1004	A003	750.00	Null
1003	A001	300.00	Null
1005	A004	900.00	Null
1006	A002	450.00	Null
1007	A005	150.00	Null

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Question 7

SQL Query

```
SELECT app_id AS application_id,
       customer_id,
       loan_type,
       amount_requested
FROM gauteng_loan_applications
UNION ALL
SELECT app_id AS application_id,
       customer_id,
       loan_type,
       amount_requested
FROM western_cape_loan_applications;
```

Results Table

application_id	customer_id	loan_type	amount_requested
LA001	C1001	NULL	NULL
LA002	C1002	NULL	NULL
LA003	C1003	NULL	NULL
LA004	C1004	NULL	NULL
LA003	C1003	NULL	NULL
LA005	C1005	NULL	NULL
LA006	C1006	NULL	NULL
LA007	C1007	NULL	NULL

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Question 8

SQL Query

```
SELECT complaint_id,
       customer_id,
       category,
       logged_date
FROM email_complaints
UNION ALL
SELECT complaint_id,
       customer_id,
       category,
       logged_date
FROM app_complaints;
```

Results Table

complaint_id	customer_id	category	logged_date
EC001	C2001	Null	Null
EC002	C2002	Null	Null
EC003	C2003	Null	Null
EC004	C2004	Null	Null
AC001	C2005	Null	Null
AC002	C2001	Null	Null
AC003	C2006	Null	Null
AC004	C2007	Null	Null

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Question 9

SQL Query

```
SELECT payment_id,
       account_id,
       amount,
       payment_date
FROM april_payments
UNION ALL
SELECT payment_id,
       account_id,
       amount,
       payment_date
FROM may_payments;
```

Results Table

payment_id	account_id	amount	payment_date
PAY001	A001	1250.00	Null
PAY002	A002	4800.00	Null
PAY003	A003	9200.00	Null
PAY004	A004	3300.00	Null
005	A001	2500.00	Null
006	A005	7600.00	Null
007	A002	5100.00	Null
008	A006	2800.00	Null

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Question 10

SQL Query

```
SELECT entry_id,
       account_id,
       entry_type,
       amount,
       entry_date
FROM debit_entries
UNION ALL
SELECT entry_id,
       account_id,
       entry_type,
       amount,
       entry_date
FROM credit_entries;
```

Results table

entry_id	account_id	entry_type	amount	entry_date
DR001	A001	Debit	Null	Null
DR002	A002	Debit	Null	Null
DR003	A003	Debit	Null	Null

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Bonus 01

=The fraud analyst should use union, because the goal is to find a unique list of customers who appeared in the January or February water lists.
⇒ If a customer appears on both lists, they should only return once.
⇒ UNION removes duplicate rows in order to produce a clean and deduplicated list the analyst needs.

Bonus 2

⇒ The auditing team should use UNION ALL, because they need to count every individual transaction that passed through the bank, that includes duplicate transactions that were recorded in two systems on the same day.
⇒ UNION ALL keeps all rows and includes duplicates to ensure that no transaction is dropped from the audit trail.
⇒ Using Union will incorrectly remove duplicates and understate the transaction count.

Bonus 3

⇒ The query failed because the two 'SELECT' statements return a different number of columns.
= The first 'SELECT' returns 3 columns (~~'accounts'~~, ('accounts_id', 'account_holder').

(11)

=> The second 'SELECT' returns 3 columns ('account_id', 'account_holder', 'city').
=> One of the rules in UNION states that both queries must return the same number of columns with compatible data types.

Correct SQL Query

```
SELECT account_id,
       account_holder,
       city
FROM branch_andhra_accounts
UNION
SELECT account_id,
       account_holder,
       city
FROM branch_rosebank_accounts;
```


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